general theory of relativity also thoroughly reshaped our view of space and time. In both Newton's conception and that of special relativity, space and time provided an unchanging stage for the events of the universe. Even though the slicing of the cosmos into space at successive moments has a flexibility in special relativity unfathomable in Newton's age, space and time do not respond to happenings in the universe. Spacetime—the loaf, as we've been calling it—is taken as a given, once and for all. In general relativity, all this changes. Space and time become players in the evolving cosmos. They come alive. Matter here causes space to warp there, which causes matter over there to move, which causes space way over there to warp even more, and so on. General relativity provides the choreography for an entwined cosmic dance of space, time, matter, and energy.

This is a stunning development. But we now come back to our central theme: What about the bucket? Does general relativity provide the physical basis for Mach's relationist ideas, as Einstein hoped it would?

Over the years, this question has generated much controversy. Initially, Einstein thought that general relativity fully incorporated Mach's perspective, a viewpoint he considered so important that he christened it *Mach's principle.* In fact, in 1913, as Einstein was furiously working to put the final pieces of general relativity in place, he wrote Mach an enthusiastic letter in which he described how general relativity would confirm Mach's analysis of Newton's bucket experiment.[17] And in 1918, when Einstein wrote an article enumerating the three essential ideas behind general relativity, the third point in his list was Mach's principle. But general relativity is subtle and it had features that took many years for physicists, including Einstein himself, to appreciate completely. As these aspects were better understood, Einstein found it increasingly difficult to fully incorporate Mach's principle into general relativity. Little by little, he grew disillusioned with Mach's ideas and by the later years of his life came to renounce them.[18]

With an additional half century of research and hindsight, we can consider anew the extent to which general relativity conforms to Mach's reasoning. Although there is still some controversy, I think the most accurate statement is that in some respects general relativity has a